

1. (4 pontos) Based on Kurlat (2020, Cap. 4)

Suppose North and South Korea are two closed economies described by the Solow model. Their technology levels don't grow, but their populations do, at the constant rates n_N and n_S . Their initial populations and capital stocks are:

$$\begin{aligned} L_{S,0} &= 20 \\ K_{S,0} &= 8000 \\ L_{N,0} &= 10 \\ K_{N,0} &= 100 \end{aligned}$$

The production function for each country $i \in \{N, S\}$ is given by

$$Y = A_i K^{\alpha_i} L^{1-\alpha_i}$$

with α and A being different between them. North Korea has $\alpha_N = 0.25$ and $A_N = 4$, while South Korea has $\alpha_S = 0.5$ and $A_S = 5$.

Each country saves a constant fraction s_i of its output and capital depreciates at the common rate $\delta = 0.05$, so that capital per worker $k_{i,t} \equiv K_{i,t}/L_{i,t}$ evolves according to

$$k_{i,t+1} = \frac{(1 - \delta)k_{i,t} + s_i A_i k_{i,t}^{\alpha_i}}{1 + n_i}$$

South Korea saves more but grows demographically more slowly than the North:

$$\begin{aligned} s_S &= 0.30 & n_S &= 0.01 \\ s_N &= 0.16 & n_N &= 0.03 \end{aligned}$$

- Compute capital per worker and GDP per capita in each country in period 0.
 - Derive the steady-state capital per worker k_i^* as a function of $(s_i, A_i, \alpha_i, \delta, n_i)$ and compute k_N^* , k_S^* , y_N^* and y_S^* . What does the model therefore predict about growth in each country?
 - Assume the unification of both countries leads to production in Korea happening with α_S and A_S , while using the stocks of capital and labor from both countries. What is the change in GDP per capita for both countries? Where does the gain come from in each case? Does unification raise aggregate Korean output?
 - (Bonus.) Suppose the unified Korea saves at rate s_S and its labor force grows at the labor-weighted average of n_N and n_S . Compute k_U^* and y_U^* . Is the unified Korea still below its steady state? Compare y_U^* with y_S^* and explain the difference.
- 2. (6 pontos) Based on Kurlat (2020, Cap. 5)**
- In the city of Gotham the production function is given by

$$Y = AK_P^\alpha L^{1-\alpha} \tag{1}$$

where K_P is capital used in production. Factor markets are perfectly competitive and the price of output is normalized to 1.

Unfortunately, crime is a significant problem in Gotham, so for each unit of productive capital, firms need to invest θ units of capital in security (i.e., cameras, armored cars, and masked vigilantes). This ratio is fixed, and security capital is recorded in the national accounts as capital, alongside productive capital. Use K to denote the *total* capital stock, as opposed to K_P , which denotes capital used in production.

Based on this setup, answer the following items. Where a numerical answer is requested, use $\alpha = 1/3$ and $\theta = 0.5$.

- a) Find an expression for total output as a function of A, K, L, θ and α . What fraction of the measured capital stock is devoted to security? How much output is lost due to the need for security?
- b) Write down and solve the problem of a firm that chooses K_P and L to maximize profits, taking w and r as given. Notice that the firm has to pay the same rental rate on security capital as on productive capital and compute the factor shares that a statistical agency would measure. Are they distorted by the presence of crime?
- c) Suppose a "riddler" studying Gotham applies the development accounting exercise of **Kurlat Section 5.3**. They observe Y , K and L accurately, but cannot separate productive from security capital pool them together. What estimate of productivity $\hat{A}$ do they obtain, and how does it compare to the true value of A ? In light of item (b), is the riddler's value of α also wrong?
- d) WayneTech invents a new security technology which reduces the security capital requirement from θ to $\theta' = \frac{1}{4}\theta$. Suppose the transition takes place over a decade, during which the total capital stock grows at g_K and the labor force at g_L . Using equation (5.4.2), write out the growth accounting decomposition of g_Y and solve for the Solow residual. True technology A has not changed at all: show that the entire effect of the new technology is reported as TFP growth, and compute the implied annual rate numerically. Explain why this is consistent with Kurlat's statement that the residual captures "changes in policies that lead to better (or worse) allocation of resources", and state what the accounting exercise would report if instead crime had worsened.